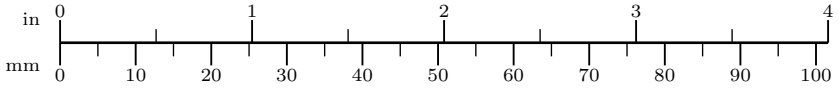
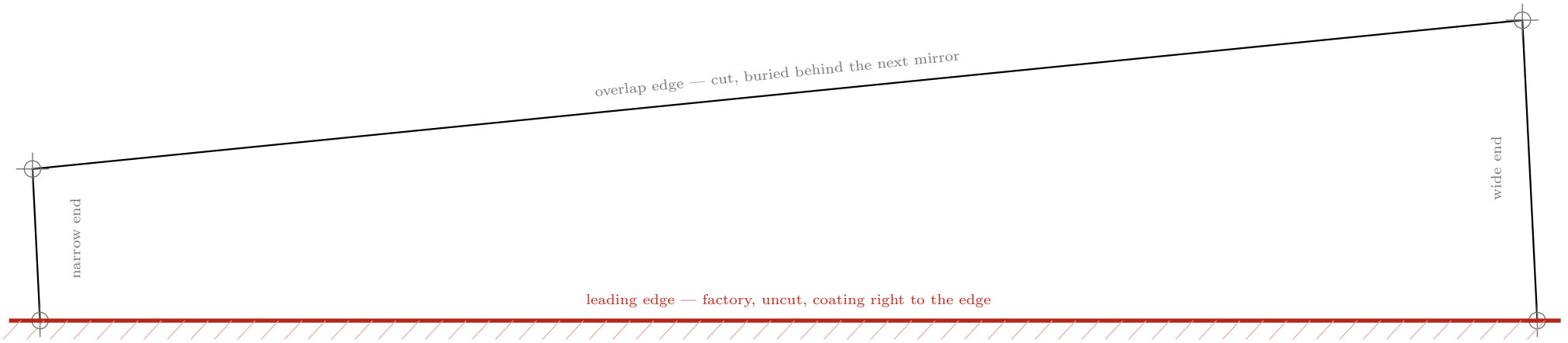
Kaleidoscope mirror cutting template

255 mm scope · one blank per sheet · drawn 1:1

Print at 100%. *Actual size*, not *Fit to page*. Check the ruler below with a real ruler first. This sheet is nearly as wide as the paper, so also check that the circled marks at *both* ends of the blank actually printed — if either is missing, your printer clipped the margins and everything here is short.



0–100 below = 100.0 mm exactly
0–4 above = 4.000 in exactly



FACTORY EDGE OF THE TILE — lay this line on it. Never cut or sand along it.

Everything below the line hangs off the tile, which is how you sight the alignment.

Blank: parallel edges **25.88** and **51.28 mm**, both slanted edges **255.42 mm**, height between the parallel edges **255.11 mm**. Needs 256.7 mm of factory edge against a 298.45 mm tile, so one blank per edge — three edges, or three tiles.

Placing it. Leave the protective film on. Lay the sheet so the heavy line sits on a factory edge, with the strip below it overhanging so you can sight the alignment, and tape it down. Prick through the centre of the four circled marks, lift the paper, join the pricks with your straightedge, and cut only the three non-factory edges.

Coated side up. These are rear-surface mirrors. A jeweller's saw cuts on the downstroke, so the teeth enter the top face and break out of the bottom; coated side up puts that breakout on the clear face. Blade: a #2, teeth pointing toward the handle.

Mark the factory edge on the coated back as soon as the blank is free. Once the other three edges are cut it is easy to lose track of which one it was, and at assembly it has to end up on the corner-block side of its plate.

Width is forgiving, and not in the way you would guess. Each mirror is pushed until its leading edge stops against its neighbour, so *that* edge lands on the vertex whatever the blank measures. Width only sets how much surplus buries itself behind the next mirror. Anything from 25.1 to 27.0 mm across the narrow edge will seat. Too wide is the direction that fights you — past the limit the surplus runs out of dead space and jams against the neighbouring plate. Too narrow only costs you: with no surplus left, the next mirror's leading edge has nothing to land on and the seam shows as a dark line. So if you are going to miss, miss narrow.

Sanding. Only the cut edges, never the factory edge: 400-grit silicon carbide to take the burr off, 600 to finish.